SOFTWARE REQUIREMENTS SPECIFICATION

Health Care Insurance Revenue Plan

Version 1.0

Customer Segmentation, Offer Targeting & Royalty Analytics Platform

Prepared by: Data & Engineering Team
Date: April 22, 2026
Status: Draft

1. Introduction

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Health Care Insurance Revenue Plan platform. The system will enable a health insurance company to collect, process, and analyze customer and claims data sourced from third-party providers in order to drive targeted marketing, loyalty rewards, and improved revenue outcomes.

The architecture of this platform follows a structured data pipeline: raw data is ingested from external sources, stored in Amazon S3, cleaned and transformed using PySpark on Amazon EMR, and ultimately surfaced as actionable intelligence through a reporting dashboard. Key deliverables include customer segmentation models, personalized offer recommendations, royalty calculations for long-term policyholders, and claim rejection analysis.

1.1 Purpose

The purpose of this SRS is to serve as the authoritative reference for all stakeholders involved in the design, development, testing, and deployment of the Health Care Insurance Revenue Plan platform. It documents the agreed-upon requirements that the system must fulfill and provides a shared understanding of scope, constraints, and success criteria.

This project will directly support the insurance company in:

- Identifying which customer segments are most likely to purchase or renew insurance policies.
- Designing and delivering targeted offers that align with each segment's health profile and financial situation.
- Rewarding loyal customers through a transparent and auditable royalty calculation system.
- Reducing revenue loss by analyzing patterns in rejected claims and surfacing corrective actions.
- Building customer trust through data-driven personalization and consistent engagement.

1.2 Intended Audience and Use

This document is intended for the following audiences:

Project Managers:

Use this SRS to scope development sprints, define milestones, track requirements traceability, and communicate progress with executive stakeholders. The product scope section and functional requirements table are especially relevant.

Software Developers / Data Engineers:

Use this SRS to understand data flows, system interfaces, and the technical requirements for building ETL pipelines, segmentation models, and the reporting dashboard. Functional requirements and system feature sections provide implementation guidance.

QA Testers:

Use this SRS as the basis for test case design. Each functional and non-functional requirement should map to one or more test cases. Pay close attention to data quality thresholds, security requirements, and performance benchmarks.

Business Analysts & Marketing Teams:

Use the overall description and functional requirements sections to understand how customer segmentation and offer targeting logic will work, and how royalty programs will be operationalized.

1.3 Product Scope

The Health Care Insurance Revenue Plan platform encompasses the following capabilities:

- Ingestion and storage of third-party health insurance data in a cloud data lake (Amazon S3).
- Automated data profiling, cleaning, and transformation using PySpark on Amazon EMR.
- Customer segmentation based on demographics, health conditions, claim history, and premium behavior.
- A recommendation engine that generates targeted insurance offer packages for each customer segment.
- A royalty calculation module that computes loyalty rewards for qualifying long-term policyholders.
- Claim rejection analysis to identify patterns, root causes, and corrective recommendations.
- Competitor data benchmarking to evaluate pricing and coverage gaps relative to market alternatives.
- A business intelligence dashboard providing KPIs, trend analysis, and exportable reports for management.

Out of scope for this version: customer-facing web portals, real-time claims processing, and integration with core policy administration systems.

1.4 Definitions and Acronyms

The following table defines all key terms, acronyms, and abbreviations used throughout this document.

Term / Acronym	Definition
SRS	Software Requirements Specification — a document that describes the intended purpose, environment, and requirements of a software system.
ETL	Extract, Transform, Load — a data integration process that extracts data from sources, transforms it into a usable format, and loads it into a target system.

S3	Amazon Simple Storage Service — a scalable object storage service in Amazon Web Services used to store raw and processed data files.
EMR	Amazon Elastic MapReduce — a cloud big-data platform on AWS for running large-scale distributed data processing jobs using frameworks like Apache Spark.
PySpark	Python API for Apache Spark — an open-source distributed computing framework used for large-scale data processing, ETL, and analytics.
Subscriber	An individual or organization enrolled in a health insurance plan and responsible for premium payments.
Claimant	A subscriber or covered dependent who submits a claim for medical services covered under an insurance policy.
Subgroup	A subdivision within a Group, used to further classify policyholders based on department, location, or coverage tier.
Group	A collection of subscribers typically from the same employer or organization sharing a common insurance policy.
Premium	The periodic payment made by a subscriber to maintain active health insurance coverage.
Royalty	A loyalty reward or financial benefit offered to long-term customers who have maintained or renewed insurance policies over time.
KPI	Key Performance Indicator — a measurable value that indicates how effectively business objectives are being met.
HIPAA	Health Insurance Portability and Accountability Act — US regulation governing the privacy and security of health information.
PII	Personally Identifiable Information — data that can be used to identify a specific individual.

2. Overall Description

This section provides a high-level description of the Health Care Insurance Revenue Plan platform, covering the context in which it will operate, the users it will serve, and the key assumptions and dependencies that underpin its design.

2.1 Product Overview

The platform is a new internal analytics and data engineering system built for a health insurance company. It is not a replacement of any existing operational system, but rather a supplementary intelligence layer that aggregates and processes data from multiple sources to enable smarter business decisions.

The system will integrate with Amazon Web Services infrastructure and will be developed using Python and PySpark. Its outputs — segmented customer lists, offer recommendations, royalty calculations, and dashboards — will be consumed by internal business teams rather than end customers.

2.2 User Needs

The following user groups have been identified, along with their primary needs:

Marketing & Product Teams

- Need to identify which customer segments are underserved or most receptive to new policy offers.
- Require automated, data-driven offer recommendations rather than manual outreach strategies.
- Want visibility into competitor pricing and coverage to refine their own offerings.

Finance & Actuarial Teams

- Need accurate royalty calculations to plan for loyalty program payouts.
- Require claim rejection trend data to model financial risk and reduce unexpected losses.
- Want consolidated revenue KPI reporting with historical trend analysis.

Data Engineering Team

- Needs clearly defined data schemas, primary/foreign key documentation, and pipeline architecture specifications.
- Requires automated data quality checks and alerting to maintain pipeline reliability.

Compliance & Legal Teams

- Need assurance that all data handling complies with HIPAA and applicable data protection regulations.
- Require comprehensive audit logs of all data access and transformation activities.

2.3 Assumptions and Dependencies

The following assumptions have been made in preparing this SRS. If any of these assumptions prove false, requirements may need to be revisited.

ID	Assumption
A-01	Third-party data providers will supply datasets in agreed-upon formats (CSV/JSON/Parquet) with documented schemas.
A-02	AWS infrastructure (S3, EMR, IAM) will be provisioned and accessible prior to development kickoff.
A-03	All required data sharing agreements and HIPAA Business Associate Agreements (BAAs) will be executed before data ingestion begins.

A-04	Source data will include at a minimum: subscriber demographics, policy details, claims records, and historical premium payments.
A-05	Downstream consumers (project managers, marketing teams) will have access to the reporting dashboard via a supported web browser.
A-06	A dedicated data engineering team with PySpark expertise will be available for development and maintenance.
A-07	Business rules for royalty calculation and offer targeting will be provided and approved by the product owner prior to implementation.

Dependencies include: AWS S3, AWS EMR, Apache PySpark, a BI visualization tool (e.g., Apache Superset or Amazon QuickSight), and version control via Git.

3. System Features and Requirements

3.1 Functional Requirements

The table below lists all functional requirements for the platform. Each requirement is assigned a unique identifier for traceability.

ID	Feature	Description
FR-01	Data Ingestion	The system shall ingest health insurance data from third-party sources (CSV, JSON, Parquet) into Amazon S3 landing zones on a scheduled or event-driven basis.
FR-02	Data Profiling	The system shall automatically profile incoming datasets to identify data types, null percentages, duplicate records, and statistical distributions.
FR-03	Primary & Foreign Key Identification	The system shall detect and document primary keys (e.g., Subscriber ID, Claim ID) and foreign key relationships across all ingested datasets.
FR-04	Data Cleaning	The system shall cleanse ingested data by removing duplicates, imputing or flagging null values, standardizing date/phone/address formats, and correcting data type mismatches.
FR-05	Data Transformation	The system shall transform raw data using PySpark on EMR into normalized analytical tables suitable for reporting and segmentation.
FR-06	Customer Segmentation	The system shall segment customers by age group, chronic disease category, claim frequency, premium tier, and geographic region.
FR-07	Offer Recommendation Engine	The system shall generate personalized insurance offer recommendations for each customer segment based on behavioral and health profile data.

FR-08	Royalty Calculation	The system shall compute royalty amounts for qualifying long-term policyholders based on tenure, claim history, and policy tier rules.
FR-09	Claim Rejection Analysis	The system shall analyze rejected claims to surface rejection reason codes, trends, and affected customer cohorts to reduce future rejections.
FR-10	Competitor Data Analysis	The system shall process and compare third-party competitor insurance data to benchmark policy pricing and coverage gaps.
FR-11	Revenue Reporting Dashboard	The system shall provide a reporting dashboard displaying KPIs such as total premiums collected, claims paid, rejection rates, and projected royalty payouts.
FR-12	Audit Logging	The system shall maintain an immutable audit log of all data ingestion, transformation, and user-access events with timestamps.

3.2 External Interface Requirements

3.2.1 User Interface

The system shall expose a web-based business intelligence dashboard accessible via a modern browser (Chrome, Firefox, Edge). The dashboard shall:

- Display summary KPI cards on the home screen (total active subscribers, claims paid YTD, rejection rate, royalties due).
- Allow filtering of all reports by date range, customer segment, geographic region, and policy type.
- Support export of reports and data tables in CSV and PDF formats.
- Provide role-based views so that marketing, finance, and compliance teams see only data relevant to their function.

3.2.2 Hardware Interfaces

The platform is cloud-native and does not depend on any specific on-premise hardware. All compute and storage resources will be provisioned on AWS. Local developer machines must meet the following minimum specifications: 8 GB RAM, a modern 64-bit processor, and a stable internet connection with access to AWS endpoints.

3.2.3 Software Interfaces

The system will interface with the following software components:

- Amazon S3: Primary data lake storage for raw, cleansed, and transformed data layers.
- Amazon EMR: Managed cluster platform for executing PySpark ETL jobs.
- AWS IAM: Identity and access management for controlling permissions to S3 buckets, EMR clusters, and dashboards.
- Apache Spark / PySpark: Distributed data processing engine for all transformation and analytical workloads.
- BI Dashboard Tool (Apache Superset or AWS QuickSight): Front-end reporting and visualization layer.

- Third-Party Data Providers: External APIs or SFTP endpoints for receiving competitor and supplementary insurance data.

3.2.4 Communications Interfaces

All data transfers between system components shall use encrypted channels (TLS 1.2 or higher). Third-party data ingestion will use SFTP or HTTPS. Internal service communication within AWS will leverage AWS VPC endpoints to avoid public internet exposure.

3.3 System Features

3.3.1 Data Ingestion & Storage Pipeline

The system shall provide an automated ingestion pipeline that monitors configured S3 landing zones or external endpoints for new data files. Upon detection, the pipeline shall validate file schemas, log receipt timestamps, and stage data for profiling. Supported formats: CSV, JSON, Parquet. Scheduling: configurable cron-based or event-driven triggers via AWS Lambda or EMR Step Functions.

3.3.2 Data Quality & Cleansing Engine

Before any analytical processing, all ingested data must pass through a quality gate that:

- Checks completeness: flags records where critical fields (Subscriber ID, Date of Birth, Policy Number) are null.
- Checks uniqueness: deduplicates records based on composite primary keys.
- Validates formats: ensures dates conform to ISO 8601, phone numbers to E.164, and ZIP codes to standard formats.
- Outputs a data quality report per dataset run with completeness scores and rejection summaries.

3.3.3 Customer Segmentation Engine

The segmentation engine shall classify customers into defined cohorts using configurable business rules and statistical clustering. Segmentation dimensions shall include:

- Age group (e.g., 18-30, 31-45, 46-60, 60+).
- Chronic disease category (e.g., diabetes, cardiovascular, respiratory, none).
- Claim frequency tier (low: 0-2 claims/year, medium: 3-5, high: 6+).
- Premium payment tier (based on monthly premium amount relative to plan cohort averages).
- Geographic region (state and metropolitan area level).
- Policy tenure (new: <1 year, mid: 1-5 years, loyal: 5+ years).

3.3.4 Offer Recommendation Engine

Based on a customer's segment profile, the system shall generate a ranked list of up to three recommended insurance products or policy upgrades. Recommendations shall be updated with each ETL run cycle. The recommendation logic shall be configurable by the product team without requiring code changes (rule configuration via a business rules table stored in S3 or a managed database).

3.3.5 Royalty Calculation Module

The royalty module shall calculate rewards for customers who meet qualifying criteria. Default qualifying criteria (configurable):

- Policy tenure of 5 or more continuous years with no lapse in premium payment.
- Fewer than 3 claim rejections in the trailing 3 years.
- Active policy status at the time of calculation.

Royalty amounts shall be computed as a percentage of annual premium paid, with tiers increasing with tenure length. All calculations shall be stored with a full audit trail showing input values, applied rules, and output amounts.

3.3.6 Claim Rejection Analysis Module

The system shall analyze historical claims data to produce:

- A frequency table of rejection reason codes, ranked by volume and financial impact.
- Trend analysis showing rejection rate changes over time by reason code and customer segment.
- A list of individual customers with high rejection rates flagged for relationship management review.
- Recommendations for policy or process changes to address the top 5 rejection reason codes.

3.4 Non-Functional Requirements

The table below defines the non-functional requirements governing system quality attributes.

ID	Category	Requirement
NFR-01	Performance	The ETL pipeline shall process up to 10 million records within 4 hours using EMR auto-scaling cluster configurations.
NFR-02	Scalability	The system shall scale horizontally to handle a 5x increase in data volume without architectural changes, using S3 partitioning and EMR dynamic resource allocation.
NFR-03	Availability	The data pipeline shall achieve 99.5% uptime on a monthly basis, excluding scheduled maintenance windows communicated at least 48 hours in advance.
NFR-04	Security	All PII and health data shall be encrypted at rest (AES-256) and in transit (TLS 1.2+). Access shall be governed by role-based access control (RBAC).
NFR-05	Compliance	The system shall comply with HIPAA privacy and security rules for all data storage, processing, and transmission activities.

NFR-06	Data Quality	Post-transformation datasets shall achieve a minimum data completeness score of 95% and a duplicate rate of less than 0.5%.
NFR-07	Usability	The reporting dashboard shall be operable by non-technical business users with minimal training, with key KPIs accessible within 3 clicks from the home screen.
NFR-08	Maintainability	All ETL pipeline code shall follow PySpark coding standards, be version-controlled in Git, and include unit test coverage of at least 80%.
NFR-09	Auditability	The system shall retain all audit logs for a minimum of 7 years to satisfy regulatory and compliance requirements.
NFR-10	Disaster Recovery	The system shall support recovery point objective (RPO) of 24 hours and recovery time objective (RTO) of 8 hours for critical data pipelines.

4. Appendix

4.1 Data Architecture Overview

The platform follows a three-layer data lake architecture on Amazon S3:

Layer 1 — Raw / Landing Zone:

Raw data files as received from third-party providers. No transformation applied. Immutable once written. Partitioned by source and ingestion date.

Layer 2 — Cleansed / Standardized Zone:

Data that has passed quality checks and been standardized in terms of formats, types, and encoding. Primary and foreign keys are documented and enforced at this layer. Stored in Parquet format for efficient querying.

Layer 3 — Analytical / Curated Zone:

Aggregated and transformed tables optimized for reporting and the recommendation/royalty engines. Includes customer segmentation tables, offer mapping tables, royalty calculation outputs, and claim rejection summaries.

4.2 Key Data Entities

The following core entities have been identified from the source data:

- Subscriber: Subscriber_ID (PK), Name, Date_of_Birth, Gender, Address, Phone, Email, Group_ID (FK), Subgroup_ID (FK).
- Policy: Policy_ID (PK), Subscriber_ID (FK), Policy_Type, Start_Date, End_Date, Premium_Amount, Status.

- Claim: Claim_ID (PK), Policy_ID (FK), Subscriber_ID (FK), Service_Date, Claim_Amount, Claim_Status, Rejection_Code.
- Group: Group_ID (PK), Group_Name, Employer, Region.
- Subgroup: Subgroup_ID (PK), Group_ID (FK), Subgroup_Name, Coverage_Tier.
- Offer: Offer_ID (PK), Segment_Code (FK), Product_Name, Discount_Rate, Validity_Period.
- Royalty: Royalty_ID (PK), Subscriber_ID (FK), Calculation_Date, Base_Premium, Royalty_Rate, Royalty_Amount.

4.3 Document Revision History

Version	Date	Author	Description
1.0	4/22/2026	Data & Engineering Team	Initial draft of SRS document.